Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “diffraction” and “wavelength” to explain your answer. Sentence starters are optional.

diffraction: wave spreading around an edge or through an opening

wavelength: distance between corresponding points on repeated waves

boundary: the place where a medium or constraint changes

01 Warm-up: bring back an earlier idea

A wave has frequency **4 Hz** and wavelength **6 m**. **Find its speed**.

Word help: frequency = number of cycles each second

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

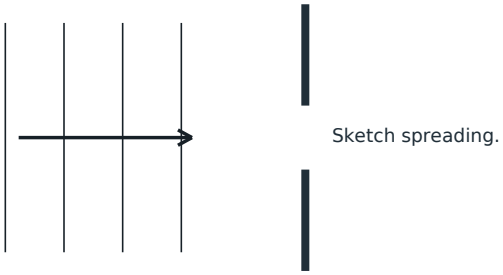
02 Worked example: explain the prediction

Reference: λ = v/f; **compare opening width** with λ. Spreading is pronounced when the opening width is comparable to wavelength. **Label the opening** and wavefronts.

Worked reasoning: A wave with wavelength **1 m** meets openings **1 m** and **10 m** wide. Spreading relative to opening size is more noticeable for the **1 m** opening.

Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.

Sentence starter: In my work, I used … to show … . I labeled it … .

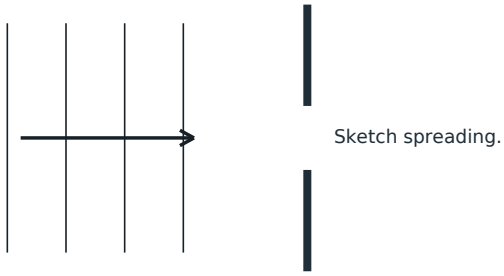


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch wavefronts after a narrow** opening about one wavelength wide.
2. Compare with a much wider opening.

Sentence starter: Both … . They differ in … because … .



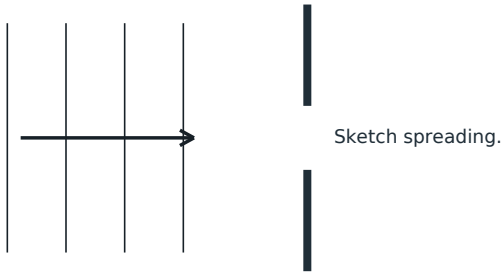
Marzano check - circle: ① ② ③

04 Apply the physics

A ripple-tank simulation sends **1 m** wavelength waves toward two openings: A is **1 m** wide and B is **10 m** wide.

1. **Sketch the emerging wavefronts**.
2. Which opening produces more noticeable spreading relative to its width?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

Diffraction

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

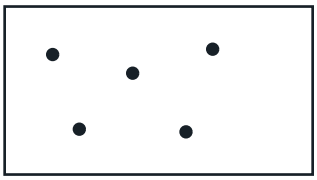
06 Review: connect an earlier idea

At the same kelvin temperature, sample B contains **4** times as many particles of the same ideal gas as A. **Compare average energy per particle** and total translational energy.
Use particle arrows and labels to **show the comparison**.

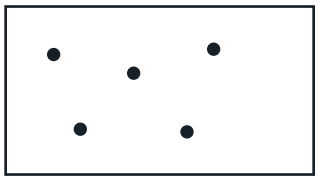
Word help: temperature = a measure of thermal state; it is not an energy amount

Sentence starter: Both … . They differ in … because … .

Sample A



Sample B



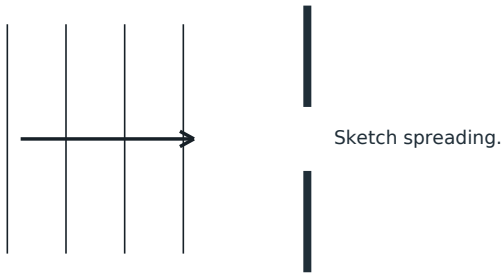
Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

07 Exit check

For the same opening, would a longer wavelength generally spread more or less?

Sentence starter: I notice … . This supports … because … .



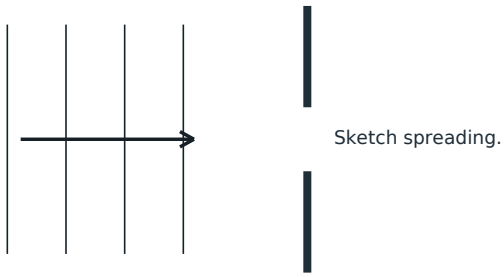
Sketch spreading.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Keep opening width fixed and halve wavelength **0**.
2. **Predict the change** in relative spreading and **draw before**/after patterns.

Sentence starter: I predict … . I used … to make this prediction because … .



Sketch spreading.

Marzano check - circle: 1 2 3